DEMKit - Quick Start for the Next Developer

Status: handover for the current development prototype.

This project has two ways to run DEMKit on Windows: the Python/Node.js developer launcher, and the internal Electron installer prototype. The installer source has been merged into main, but release testing is not finished.

Find the project

- Repository: https://github.com/kiarashdelavar/demkit-microgrid-simulator
- Technical setup: Configuration guide
- Architecture: System architecture
- Windows packaging: Installer build and test guide
- Prior tests: Sprint 1 test results
- Backlog: Product backlog

Run the development version on Windows

From the repository root, with Python 3.12, Node.js and the project dependencies already installed:

1. Follow the configuration guide to create .venv, install requirements.txt, install workspace/touchtable/web dependencies, and copy the portable example config to demkittt/conf/usrconf.py.
2. Close old DEMKit servers using ports 3001 and 3002.
3. Start both processes with:

```
.\workspace\touchtable\touchtafel.bat
```

4. The launcher checks the backend and web server, then opens http://localhost:3002.
5. Press Ctrl+C in the launcher terminal to stop the processes.

To check prerequisites without starting the app:

```
.\.venv\Scripts\python.exe .\scripts\launch_demkit.py --check
```

Run the launcher tests:

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -p "test_launcher.py"
```

Build the internal Windows prototype

This is a developer-only build; don't publish the result.

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -p "test_windows_packaging.py"
```

```
.\.venv\Scripts\python.exe .\scripts\build_windows_prototype.py
```

Read the full build guide before using these commands. The packaged backend is made with PyInstaller; Electron contains a Node.js runtime and displays the existing browser game.

The builder writes the output under packaging/windows/dist/. The installer was confirmed to work on the original Windows development computer, including reopening from the desktop shortcut and interacting with devices. That does not establish that it works on a clean computer.

What needs to be tested next

- BK-10: clean Windows x64 computer or VM without Python/Node.js installed; startup, devices, shutdown, uninstall, and remaining application data.

- BK-10: review third-party licences/redistribution permission, inherited web dependencies and signing before student distribution.

- BK-11 and BK-12: investigate and validate macOS and Linux separately; the Windows .exe does not work there.

- BK-13: ask a teacher/first-time user to follow the teacher guide and collect feedback.

The simulation has a fixed 1920 x 1080 coordinate system. A scrolling change was tested and merged, but an already-installed application only receives that change after rebuilding and reinstalling the desktop package.

Known issues / practical limits

- InfluxDB logging is postponed; legacy connection warnings may appear even while the game works.

- The model dynamically loads Python components. The first frozen build failed because components. could not be imported; the packaging entry now adds the correct parent directory to sys.path.

- If the frozen backend stops before startup, read the error log path from Electron’s dialog or run the packaged backend directly as described in the Windows packaging guide.

- If an old port stays occupied, first identify it. Do not kill every Python or Node.js process on the user’s computer.

- The inherited game does not model all real-world conditions, costs or unequal access to energy technology.

- Whether progress survives a full application restart is not a guaranteed feature.

Before handing over

Run the full simulation, make sure EV, solar and battery interactions work, and add a short note to the relevant GitHub issue describing the environment, what was tested and any actual errors. Never commit demkittt/conf/usrconf.py, credentials, built binaries or user-data folders.

Project scope: Make the existing simulator easier to install and teach with. Keep the energy engine unchanged unless a specific, verified bug requires it.